

advancements surpass previous models in terms of performance and application scope, expanding signal generation capabilities across various fields.

Compactness with embedded reflectometer and advanced modulation

Network and design engineers often need more bench space for extra test equipment to achieve higher frequencies, greater bandwidths, and complex modulation schemes. The Keysight N5186A MXG addresses this by simplifying setups with fewer external connections and up to four channels in a compact 2U form factor, cutting space requirements by 75% compared to traditional setups. As the first signal generator with an embedded reflectometer, it delivers highly accurate signals to the device under test (DUT). This optimisation maximises equipment density without compromising performance. Constructed with durable cast aluminium, these devices are suitable for both lab and field use. Their compact, handheld design ensures portability and adaptability to diverse workspaces. User-friendly interfaces simplify setup and operation, catering to users of all skill levels. Additionally, these signal generators offer optional features like pulse modulation and external frequency sweep triggers, seamlessly integrating with various platforms and programming environments for a wide range of testing applications.

The integration of embedded reflectometers and compact multi-channel designs not only minimises the physical footprint of testing setups but also simplifies external connections. Enhanced signal accuracy and testing precision streamline processes by eliminating the need for certain external measurement tools, resulting in

Future scope of technical advancements

- **Higher frequencies and bandwidths.** Development of signal generators capable of operating at higher frequencies and wider bandwidths will support emerging communication standards, such as 6G and advanced radar systems.
- **Improved accuracy and stability.** Enhanced precision and stability in signal generation will be critical for applications requiring high fidelity, such as quantum computing and sensitive scientific measurements.
- **AI and machine learning integration.** Incorporating AI and machine learning algorithms will optimise signal generation processes, enabling adaptive and real-time signal adjustments based on environmental conditions and specific application needs.
- **Miniaturisation and portability.** Advances in semiconductor technology will lead to smaller, more portable signal generators without compromising performance, making them ideal for field applications and mobile testing.
- **Enhanced software capabilities.** Future software enhancements will offer more intuitive user interfaces, automated calibration, and improved simulation tools, facilitating faster development cycles and reducing the need for extensive manual intervention.

These advancements will drive innovation in various industries, from telecommunications and aerospace to medical technology and consumer electronics.

faster setup times and increased productivity in environments requiring rapid development and high precision. In telecommunications and radar systems, the latest multichannel vector signal generators, now fitting within a sleek 2U rack, can generate signals up to 8.5GHz with a modulation bandwidth of 960MHz per channel, crucial for supporting advanced wireless communications and radar systems.

Optional I/Q modulation inputs enable advanced digital modulation, allowing modulation of carriers from 400MHz to the model's maximum frequency. An internal noise generator enhances signal complexity and testing versatility. These generators are primed for emerging technological demands, supporting sophisticated techniques like multiple input multiple output (MIMO), beamforming, and multiplexing, ensuring continued relevance and effectiveness in wireless and radar technology advancements. They support various modulation techniques, including AM, FM, QM, PM, and frequency sweeps, with a standard high-quality oven-controlled crystal oscillator (OCXO) providing exceptional frequency stability and durability,

upgradable to a rubidium oscillator for high-precision applications.

Enhancing precision and versatility

The integration of rational approximation frequency synthesis (RAFS) in the latest signal generator models is a significant advancement in frequency synthesis technology. RAFS enables ultra-high-frequency resolution down to 1µHz by utilising smaller integer divisors within a phase-locked loop system, achieving unparalleled accuracy in frequency tuning. Not only does RAFS provide precise frequency control, it also significantly reduces spurious outputs, crucial for sensitive applications spanning from scientific research to sophisticated defence systems, where even minor inaccuracies can lead to significant deviations in results.

Engineered to meet diverse technological requirements, these signal generators offer a range of frequency outputs tailored to various applications. The new R&S SMB100B microwave signal generator from Rohde & Schwarz is available with four frequency options: 8kHz to 12.75GHz, 20GHz, 31.8GHz, and 40GHz. It